MYSQL AND PHPMYADMIN

Group 5

INTRODUCTION TO MYSQL

WHAT IS MYSQL?

- **MySQL** is an open-source **Relational Database Management System (RDBMS)** that is used to store, organize, retrieve, and manage data.
- As a relational database system, MySQL organizes information into **tables.** These tables can be connected to one another through relationships, allowing related information to be stored and managed efficiently.
- MySQL uses **SQL (Structured Query Language)** to communicate with the database. Through SQL commands, users can create databases and tables, add new information, search for existing records, modify data, and remove information when necessary.

EXAMPLE USES OF MYSQL

MySQL can be used to manage information for:

- Student records
- Employee information
- Customer accounts
- Product inventories
- Hospital records
- Online shopping systems
- Websites and web applications

IMPORTANT TERMS IN MYSQL

To understand MySQL, it is important to know some basic database terms

- **Database** - A database is an organized collection of related information. It provides a structured place where data can be stored and managed.
- **Table** - A table is a database structure where actual records are stored. It organizes information into rows and columns.
- **Row** - A row represents one complete record in a table. For example, one row in a student table may represent one student.
- **Column** - A column represents a specific category or field of information. For example, a student table may have columns for student ID, name, age, and course.
- **SQL** - SQL, or Structured Query Language, is the language used to communicate with and manage relational databases.
- **DBMS** - A Database Management System (DBMS) is software that allows users to create, store, organize, retrieve, and manage data in databases.

EXAMPLE OF A MYSQL TABLE

A MySQL database stores information inside tables.

students

student_id	name	age	course
1	Luis	20	BSIT
2	Jessa	19	BSIT
3	Carl	21	BSIS

This structure makes it easier to organize information and retrieve specific records when needed.

In this example:

- students is the name of the **table.**
- student_id, name, age, and course are **columns.**
- Each horizontal entry is a **row or record.**
- Each row contains information about one student.

IMPORTANT FEATURES OF MYSQL

1. **Relational Database System**

   MySQL organizes data into tables and allows relationships to be established between related tables. This makes it suitable for systems that contain different types of connected information.

2. **Open Source**

   MySQL is available as open-source software, allowing it to be widely used for learning, development, and various applications.

3. **SQL Support**

   MySQL uses Structured Query Language to allow users and applications to interact with databases. SQL can be used to create, retrieve, modify, and delete data.

IMPORTANT FEATURES OF MYSQL

4. Fast and Efficient

MySQL is designed to process database operations efficiently. It can handle large amounts of information and numerous database requests.

5. Secure

MySQL provides mechanisms for managing users, authentication, and privileges. These features help control who can access or modify database information.

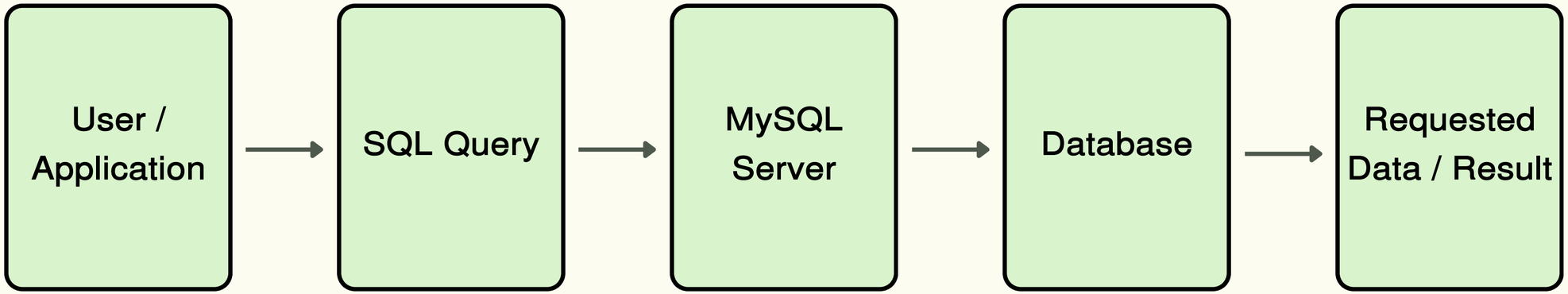
6. Widely Used

MySQL is widely used in websites, web applications, business systems, and other software applications that require structured data storage.

HOW MYSQL WORKS

- MySQL works as a **database server** that receives requests from users or applications and processes those requests.
- A user does not normally interact directly with the stored data. Instead, the user or application sends an **SQL query** to the MySQL server.

Basic Process:



EXAMPLE:

If a user wants to display all students, the following SQL command can be used:

```
SELECT * FROM students;
```

MySQL receives the query, searches the students table, and returns the records requested by the user.

INTRODUCTION TO PHPMYADMIN

WHAT IS PHPMYADMIN?

- phpMyAdmin is a free and open-source web-based tool used to administer and manage MySQL databases through a graphical user interface.
- It provides a more visual and convenient way of working with databases. Instead of relying entirely on a command-line interface, users can perform many database operations using menus, buttons, forms, and other graphical controls.
- phpMyAdmin can also be used to write and execute SQL commands, making it useful for both beginners who prefer a graphical interface and users who need to work directly with SQL.

PHPMYADMIN CAN BE USED TO:

- Create databases
- Create tables
- Insert data
- Edit data
- Delete data
- Run SQL queries
- Create relationships
- Manage database users and privileges
- Import and export databases

HOW PHPMYADMIN HELPS USERS

- phpMyAdmin provides a graphical interface that makes database management easier to understand and perform.
- For example, instead of manually writing an SQL command to create a table, a user can enter the table name, specify the number of columns, select data types, and configure settings using the interface.
- It also allows users to view existing databases and tables, inspect records, and make changes to stored information.
- However, phpMyAdmin does not replace MySQL. It works as a tool that allows users to interact with and administer MySQL databases more conveniently.

MYSQL VS. PHPMYADMIN

Although MySQL and phpMyAdmin are often used together, they are not the same thing

MySQL	phpMyAdmin
Database management system	Database administration tool
Stores and manages databases	Provides an interface for managing databases
Uses SQL	Allows users to execute SQL
Runs as a database server	Runs through a web browser
Handles database operations	Makes database management easier

ADVANTAGES OF USING PHPMYADMIN

Why use phpMyAdmin?

- **Easy to Use -** phpMyAdmin provides a graphical interface, so users can perform many database operations without having to memorize every SQL command.
- **Graphical Interface -** Databases, tables, columns, and records can be viewed through an organized visual interface, making it easier to understand the structure of a database.
- **SQL Editor -** Users can still write and execute SQL commands directly. This makes phpMyAdmin useful for practicing and testing SQL statements.
- **Database Management -** Users can create and modify databases, tables, columns, indexes, relationships, and other database components.
- **Import and Export -** Database information can be imported from or exported to files, which can be useful for transferring or backing up database information.
- **Useful for Learning -** Because it provides both graphical controls and SQL functionality, phpMyAdmin can help beginners understand how database operations work.

WHAT IS DATABASE?

- A database is an organized collection of data that can be easily accessed, managed, and updated.
- A database provides a structured way of storing information instead of keeping related information in separate, unorganized files.
- For example, a school system can use one database to organize information about students, teachers, subjects, courses, and enrollment records.

Example: School Database

A database named:

- school

may contain tables such as:

- students
- teachers
- subjects
- enrollments
- grades

CREATING A DATABASE USING SQL

A database can be created using the CREATE DATABASE statement.

Syntax:

- CREATE DATABASE database_name;

  The CREATE DATABASE command tells MySQL to create a new database using the name specified by the user.

Example:

- **CREATE DATABASE** school;

  - This creates a database named school.

  - After creating a database, it needs to be selected before tables and other objects can be created inside it.

CREATING A DATABASE USING SQL

- USE school;
- - This creates a database named school.
  - After creating a database, it needs to be selected before tables and other objects can be created inside it.

- SHOW DATABASES;
  - This command displays the databases available on the MySQL server.

CREATING A DATABASE IN PHPMYADMIN

A database can also be created through the phpMyAdmin graphical interface.

General Steps:

1. **Open phpMyAdmin**

   Launch phpMyAdmin through a web browser.

2. **Go to the Databases section**

   The Databases section provides options for creating and managing databases.

3. **Enter the database name**

   Type the desired name of the new database.

4. **Select the appropriate collation if needed**

   Collation determines how text data is compared and sorted.

5. **Click Create**

   phpMyAdmin sends the necessary request to MySQL to create the database.

6. **Open the new database**

   The newly created database will appear in the database list and can then be used to create tables.

MYSQL VS. PHPMYADMIN

Creating a Table

After creating a database, tables can be created to store actual records.

Example:

```
CREATE TABLE students (
    student_id INT,
    name VARCHAR(100),
    age INT,
    course VARCHAR(100)
);
```

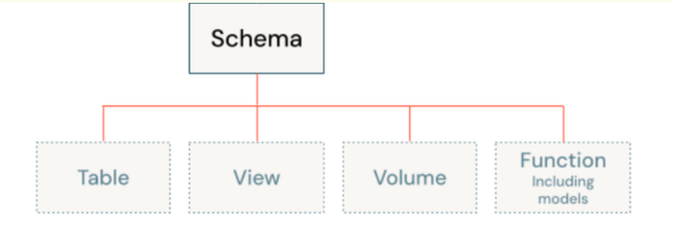
Table Structure:

Column	Data Type
student_id	INT
name	VARCHAR(100)
age	INT
course	VARCHAR(100)

DATABASE OBJECTS

DATABASE OBJECTS

- **A DATABASE OBJECT IS ANY DEFINED OBJECT IN A DATABASE THAT IS USED TO STORE OR REFERENCE DATA.**



SCHEMA

CREATE SEQUENCE sequence
[INCREMENT BY n]
[START WITH n]
[{MAXVALUE n | NOMAXVALUE}]
[{MINVALUE n | NOMINVALUE}]
[{CYCLE | NOCYCLE}]
[{CACHE n | NOCACHE}];

SEQUENCE

Name	Null?	Type
DEPTNO		NUMBER(2)
DNAME		VARCHAR2(14)
LOC		VARCHAR2(13)

Table Schema

TABLE

CREATE INDEX emp_last_name_idx
ON employees(last_name);

INDEX

ID_NUMBER	NAME	ANN_SALARY
124	Mourgos	69600
141	Rajs	42000
142	Davies	37200
143	Matos	31200
144	Vargas	30000

VIEW

CREATE SYNONYM d_sum FOR
dept_sum_vu;

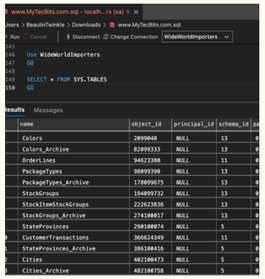
SYNONYM

DATABASE OBJECTS



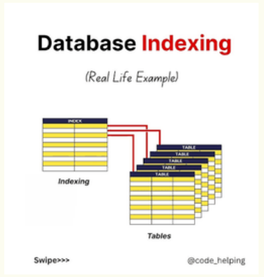
SCHEMA

A **SCHEMA** is the top-level container that holds other database objects. In MySQL, the terms database and schema are essentially interchangeable.



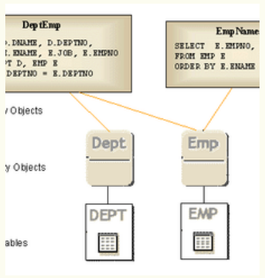
TABLE

a **TABLE** is a set of data elements (values) using a model of vertical columns (identifiable by name) and horizontal rows, the cell being the unit where a row and column intersect.



INDEX

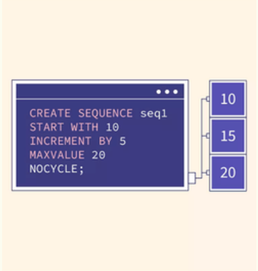
An **INDEX** is a copy of selected columns of data, from a table, that is designed to enable very efficient search.



VIEW

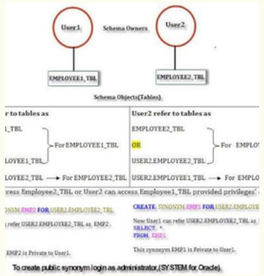
In a database, a **VIEW** is the result set of a stored query that presents a limited perspective of the database to a user.

DATABASE OBJECTS



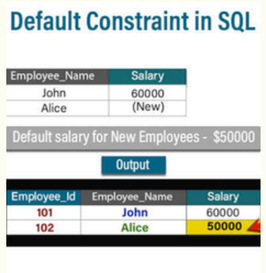
SEQUENCE

A **Sequence** is a user created database object that can be shared by multiple users to generate unique integers.



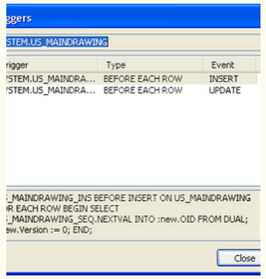
SYNONYM

- A **Synonym** alternate names for a table, view, sequence or other object in a database. Creating a synonym eliminates the need to qualify the object name with the schema.



CONSTRAINT

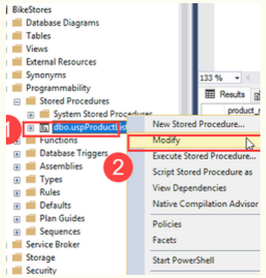
A **Constraint** is a declarative rule applied to a table column that prevents invalid data from entering the database management system.



TRIGGER

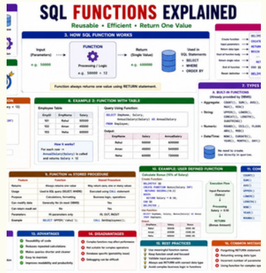
A **Trigger** is an automated script that fires instantly in response to specific data modifications like inserting, updating, or deleting table records.

DATABASE OBJECTS



STORED PROCEDURE

A **Stored Procedure** is a precompiled collection of SQL statements that executes complex business logic directly on the database server to reduce traffic.



FUNCTION

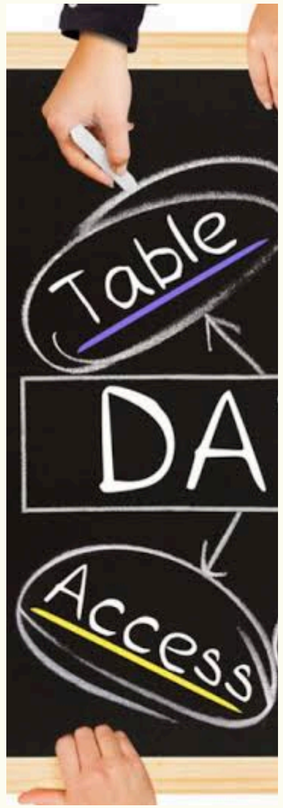
A **Function** reusable code block that accepts input parameters, performs calculations, and always returns a specific value or table.

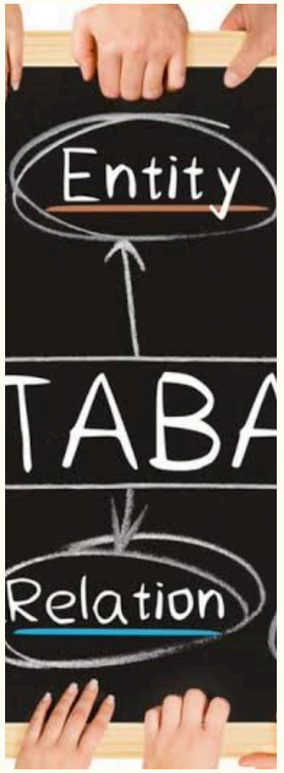
SUMMARY

Database objects are the essential building blocks that transform a raw storage repository into a secure, high-performance relational database.

Using them effectively provides three critical benefits to any application:

- Data Integrity & Consistency: Constraints and sequences act as automated gatekeepers. They prevent corrupted data from entering your system and ensure that relationships between tables remain structurally sound.
- **Performance & Scalability:** Indexes and views optimize how data is accessed and read. They dramatically speed up query response times and reduce the physical hardware strain on your server as your data grows.
- **Security & Maintainability:** Stored procedures, functions, and synonyms abstract complex logic away from the frontend. This hides your underlying table architecture from unauthorized users and allows developers to update business rules in a single, centralized location.







SQL OVERVIEW

WHAT IS MYSQL?

- **SQL stands for Structured Query Language.**
- It is the language used to communicate with relational database systems such as MySQL.
- SQL allows users to give instructions to the database. These instructions can involve creating database structures, retrieving information, adding new records, changing existing records, or removing records.

SQL CAPABILITIES & MYSQL

- Create databases and tables
- Insert data
- Retrieve data
- Update existing data
- Delete data
- Define relationships
- Control access
- Manage database structures

SQL is therefore an important part of working with MySQL because it provides the commands used to interact with stored data.

COMMON SQL COMMANDS

Some commonly used SQL commands include:

Command	Purpose
CREATE	Creates a database or object
ALTER	Changes an existing object
DROP	Removes an object
SELECT	Retrieves data
INSERT	Adds data
UPDATE	Modifies data
DELETE	Removes data

Example:

```
SELECT * FROM students;
```

This retrieves all columns and records from the students table.

The SELECT command is one of the most commonly used SQL commands because it allows users to retrieve information stored in the database.

SQL CATEGORIES

SQL commands can be grouped according to their purpose.

DQL — Data Query Language

Used to retrieve information from the database.

Example:
`SELECT`

DML — Data Manipulation Language

Used to add, modify, and remove records stored in database tables.

Examples:
`INSERT, UPDATE, DELETE`

DDL — Data Definition Language

Used to define and modify the structure of database objects.

Examples:
`CREATE, ALTER, DROP`

TCL — Transaction Control Language

Used to manage database transactions.

Examples:
`COMMIT, ROLLBACK`

DCL — Data Control Language

Used to manage access and permissions for database users.

Examples:
`GRANT, REVOKE`

SQL SYNTAX

WHAT IS SQL SYNTAX?

SQL syntax refers to the rules and structure that should be followed when writing SQL statements.

Just like programming languages have rules for writing commands, SQL also has a specific structure that allows MySQL to understand what the user wants the database to do. A correctly written SQL statement can be processed by MySQL, while an incorrectly written statement may result in a syntax error.

Basic SQL Statement:

```
SELECT column_name
FROM table_name;
```

This statement tells MySQL to retrieve specified columns from a table.

Example:

```
SELECT name
FROM students;
```

This statement tells MySQL to retrieve the name column from the students table.

SELECT STATEMENT

The SELECT statement is used to retrieve information from one or more database tables.

Select Specific Columns

```
SELECT name, age
FROM students;
```

This retrieves only the name and age columns.

Select All Columns

```
SELECT *
FROM students;
```

The * symbol means all columns.

With a Condition

```
SELECT *
FROM students
WHERE age = 20;
```

This retrieves the records of students whose age is 20. The WHERE clause is useful when users do not want to retrieve every record in a table.

INSERT, UPDATE, AND DELETE STATEMENTS

INSERT Statement

The INSERT statement is used to add new records to a table.

```
INSERT INTO students (student_id, name, age)
VALUES (1, 'Juan', 20);
```

This adds a new student record to the students table.

UPDATE Statement

The UPDATE statement is used to modify information that already exists.

```
UPDATE students
SET age = 21
WHERE student_id = 1;
```

This changes the age of the student whose ID is 1.

DELETE Statement

The DELETE statement is used to remove records from a table.

```
DELETE FROM students
WHERE student_id = 1;
```

This removes the record whose student_id is 1.

Always be careful when using UPDATE and DELETE. If a WHERE condition is not included when it is needed, multiple or all records may be changed or deleted.

SQL CLAUSES

SQL statements can contain different clauses that provide additional instructions about how data should be retrieved or organized.

WHERE

Filters records based on a condition.

```
WHERE age > 18
```

ORDER BY

Sorts the retrieved results (e.g., by name column).

```
ORDER BY name;
```

GROUP BY

Groups records for calculations or summaries.

```
GROUP BY course;
```

LIMIT

Limits the number of records returned.

```
LIMIT 10;
```

DATA TYPES

WHAT ARE DATA TYPES?

A data type defines the kind of value that a database column can store.

When creating a table, each column is assigned a data type. This tells MySQL what type of information should be stored in that particular column.

Choosing an appropriate data type helps ensure that information is stored correctly and efficiently.

MAIN DATA TYPE CATEGORIES

Numeric data type - Numeric data types are used to store numbers

What does it do?

Numerical data types are commonly used for calculations, such as addition, subtraction, multiplication, division, and comparisons

MAIN DATA TYPE CATEGORIES

int - NT is used to store whole numbers without decimal values.

age INT

Examples:

18

20

25

DECIMAL is used for exact decimal values. It is commonly used for values such as prices and other numbers where exact decimal representation is important.

price DECIMAL(10,2)

Examples:

150.50

99.99

FLOAT is used for approximate decimal numbers.

temperature FLOAT

The choice between numeric data types depends on the type of numerical information that needs to be stored.

MAIN DATA TYPE CATEGORIES

Numeric data type - Numeric data types are used to store numbers

What does it do?

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MAIN DATA TYPE CATEGORIES

String and Text Data Types:
String and text data types are used to store characters and written information.

examples:
String name = ”Mommy Oni”;
String message = ”Hello, World!”;
String address = ”Payamansion”;

char: CHAR stores fixed-length strings.
code CHAR(5)
It is useful when values are expected to have a consistent length.

example:
char grade = ’A’;
char gender = ’M’;
char symbol = ’#’;
char number = ’5’;

Varchar stores variable-length strings. name VARCHAR(100)
It is commonly used for:
Names
Addresses
Email addresses
Titles

WHAT ARE DATA TYPES?

TEXT is used for longer pieces of text.

description TEXT

It can be useful for storing descriptions or other text that may contain more characters than normally needed for a short VARCHAR field.

date and time

DATETIME stores both a date and a time.

created_at DATETIME

Example:

2026-09-17 14:30:00

bolean data type

BOOLEAN is used for logical values such as true or false.

is_active BOOLEAN

In MySQL, BOOLEAN/BOOL is treated as a synonym for TINYINT(1), commonly using:

1 = TRUE

0 = FALSE

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CONSTRAINTS

WHAT ARE CONSTRAINTS?

Constraints

Are rules that can be applied to columns in a database table.

They are used to control the type and validity of data that can be entered into a table.

What does it do?

Constraints help maintain data integrity, accuracy, and consistency.

For example, if every student should have a unique student ID, a PRIMARY KEY constraint can be used to prevent duplicate identifiers.

PRIMARY KEY

A PRIMARY KEY is used to uniquely identify each record in a table.
Example:
student_id INT PRIMARY KEY
Example table:
student_id name
1 Juan
2 Maria
3 Pedro

WHAT ARE CONSTRAINTS?

It uniquely identifies records.
It cannot contain NULL.
A table has one primary key definition, which can consist of one or multiple columns.
A primary key is important because it provides a reliable way to distinguish one record from another.

WHAT ARE CONSTRAINTS?

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Are rules that can be applied to columns in a database table.

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FOREIGN KEY

A FOREIGN KEY is used to establish a relationship between two tables.
It connects a column in one table to a key column in another table
Example:
Students
student id name
1 Juan
2 Maria

enrollments
enrollment_id student_id subject
101 1 Database
102 2 Programming

The student_id in the enrollments table can reference the student_id in the students table.

NOT NULL

The NOT NULL constraint requires a column to contain a value.
name VARCHAR(100) NOT NULL
This means the name field cannot be left as NULL.

UNIQUE

The UNIQUE constraint prevents duplicate values in a column or group of columns.
email VARCHAR(100) UNIQUE
This helps ensure that the same email value is not entered into multiple records where uniqueness is required.

CHECK

The CHECK constraint requires a value to satisfy a specified condition.
age INT CHECK (age >= 18)
This applies the specified condition to the age value.
It can be used to help prevent values that do not meet a required rule from being stored.

DEFAULT

The DEFAULT constraint provides a value automatically when no value is supplied for a column.
status VARCHAR(20) DEFAULT ’Active’
If a new record does not specify a status, the default value can be Active.

WHAT EACH PART DOES

Definition Purpose
student_id INT - Stores a whole-number student ID
PRIMARY KEY - Uniquely identifies the student
name VARCHAR(100) - Stores the student’s name
NOT NULL - Requires a name
email VARCHAR(100) - Stores the student’s email
UNIQUE - Prevents duplicate values
age INT - Stores the student’s age
CHECK - Applies a condition to the age status
VARCHAR(20) - Stores the student’s status.
DEFAULT ’Active’ - Provides a default status

SUMMARY

SUMMARY

MySQL

MySQL is a relational database management system used to store, organize, retrieve, and manage structured information.

phpMyAdmin

phpMyAdmin is a web-based graphical tool that provides an easier way to administer and interact with MySQL databases.

Creating a Database

A database can be created using SQL commands such as CREATE DATABASE or through the graphical interface provided by phpMyAdmin.

Database Objects

Database objects are components used to store, organize, retrieve, and manage information. Examples include tables, views, indexes, stored procedures, and triggers.

SUMMARY

SQL

SQL is the language used to communicate with relational databases. It can be used to create structures and manipulate or retrieve data.

SQL Syntax

SQL syntax provides the rules and structure for writing commands that MySQL can understand and execute.

Data Types

Data types specify what kind of information can be stored in a particular column, such as numbers, text, dates, times, or Boolean values.

Constraints

Constraints are rules applied to database columns to help maintain accurate, consistent, and valid data.